Research on the stability of financial network

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Abstract

In the China interbank market, the transaction behaviors of banks are similar and the interbank borrowing is short of regulation, which may result in a big problem. In this case, when the bank system faces liquidity shocks, it is difficult for the bank system to resolve the crisis only with the help of the interbank market. Therefore, how to develop monetary policies by the central bank to ease the system risk is a big issue. Considering the issue of system risk, the present paper constructs a dynamical bank network model based on agents' behavior with the method of the agent-based computational finance, and then studies the effects of the central bank on the stability of the bank system. The results show that the central bank can eliminate financial panic and achieve stability and security of the bank system with a range of appropriate monetary policy tools. Furthermore, when the bank system becomes unstable due to the short of regulation, it is difficult for the central bank to improve the stability of the bank system by adjusting the money policy of the deposit reserve rate.

Keyword: Agent-based computational finance, Stability of bank network systems, Interbank market, System risk

1 Introduction

The recent financial crisis has shown that the bank network system is very complex. The insolvency of the US investment bank Lehman Brothers in September 2008 marked a freeze in interbank markets, which forces central bank to undertake unprecedented measures to ensure liquidity provision within the interbank network system [1]. Interbank networks have the property described as a knife-edge, or robust-yet-fragile, which exhibits that the connections between banks lead to an enhanced liquidity allocation, however increase the risk of contagion in the interbank networks. The reference[2] shows that highly interconnected bank network systems are less prone to bank-runs. The fragility of an interconnected financial system was analyzed by the reference[3], which shows that the risk of system crises is increased with increasing connectivity. The massive intervention of central bank at the height of the financial crisis furthermore raises the question whether central bank interventions can effectively stabilize interbank markets and ensure banks' liquidity provision. The present paper studies the case that when the interbank market is short of liquidity, what control measures should the central bank take to ensure the stability of the bank network system. Three control measures are taken by the central bank, which are the central bank loan interests rate, investment mortgage rate, and deposit reserve rate.

Two key results are obtained. First, when the interbank markets are short of liquidity, central bank's control measures can stabilize the interbank markets subject to the connectivity of bank networks. When the connectivity of bank networks is smaller than 0.01, the three control measures of the central bank can stabilize the interbank markets, however, when the connectivity of bank networks is larger than 0.01, the effect of central bank's control measures can't work effectively. Second, comparison among the three control measures, when the connectivity is low, the best control measures is to increase the rate of investment mortgage rate.

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